

Binary Search Tree

Binary Tree $\neq$ Binary Search Tree

$Te = O(n)$

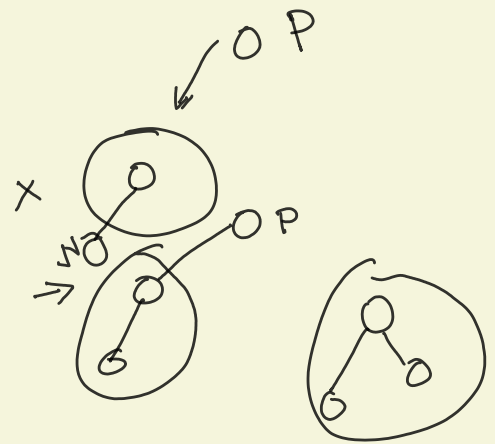
$O(\log n) \parallel \text{height} = \log n$

code
→ insertion
→ search
→ delete
→ find height

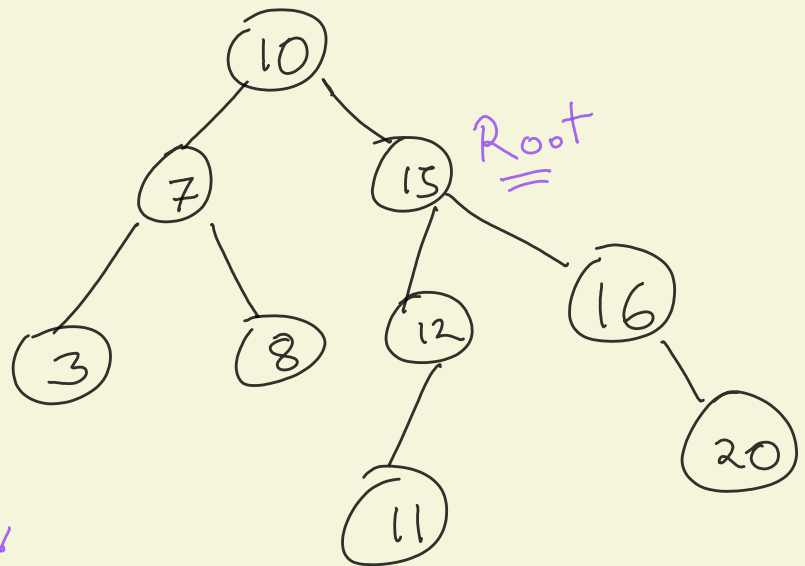
- ① print elements of BST in incremental order
- ② how to check if a BST
- ③ Inorder Traversal

Delete a node in BST

- ① no child // leaf } same
- ② one child
- ③ two child



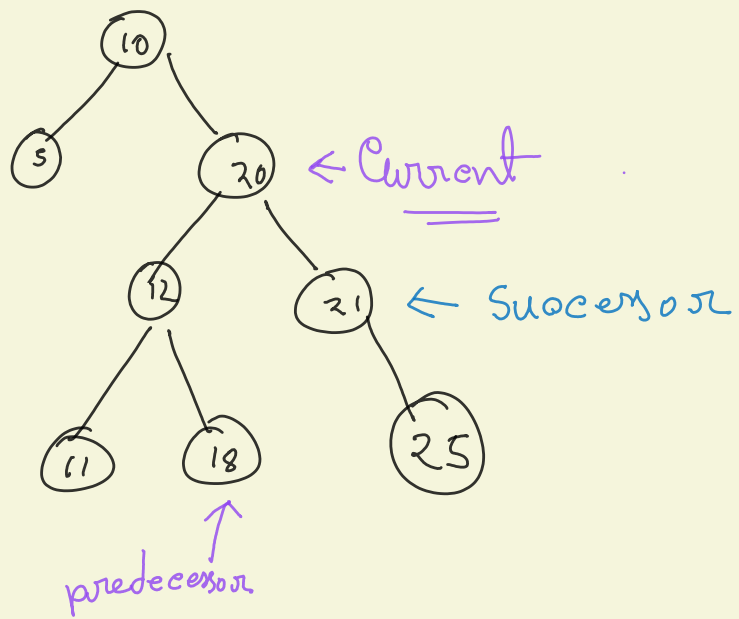
example,



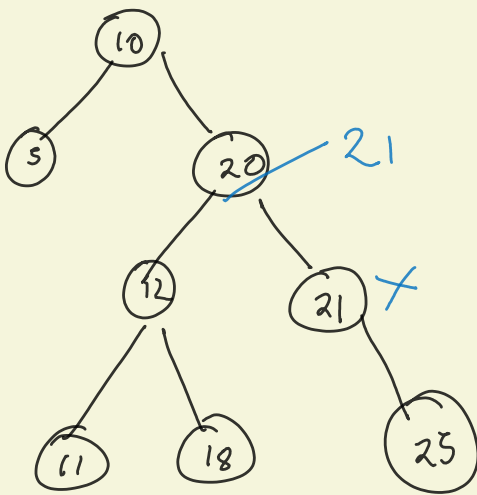
Inorder Traversal:-
3, 7, 8, 10, 11, 12, 15, 16, 20

Inorder Successor :- one step on right, extreme left
Inorder Predecessor :- one step on left, extreme right

Case 3:- node with 2 children



Successor



predecessor

